Homework Assignment 3

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1 problem 1: energy-momentum-stress tensor

for this question, its helpfull to first consider the components of $\mathbb{T}^{\mu\nu}$. Firstly consider the Einstein summation of $F^{\mu\alpha}F^{\nu}_{\alpha}$ which creates a tensor we will define as $H^{\mu\nu}$ whose components are:

$$H^{\mu\nu} = \begin{bmatrix} E^2/c^2 & \frac{1}{c}(E_yB_z - E_zB_y) & \frac{1}{c}(E_zB_x - E_xB_z) & \frac{1}{c}(E_xB_y - E_yB_x) \\ \frac{1}{c}(E_yB_z - E_zB_y) & -\frac{1}{c^2}E_x^2 + B_z^2 + B_y^2 & -\frac{1}{c^2}E_xE_y - B_xB_y & -\frac{1}{c^2}E_xE_z - B_xB_z \\ \frac{1}{c}(E_zB_x - E_xB_z) & -\frac{1}{c^2}E_xE_y - B_xB_y & -\frac{1}{c^2}E_y^2 + B_x^2 + B_z^2 & -\frac{1}{c^2}E_yE_z - B_yB_z \\ \frac{1}{c}(E_xB_y - E_yB_x) & -\frac{1}{c^2}E_xE_z - B_xB_z & -\frac{1}{c^2}E_yE_z - B_yB_z & -\frac{1}{c^2}E_z^2 + B_x^2 + B_y^2 \end{bmatrix} \tag{1}$$

Which we can divide into two regions, looking at figure 1:

$$H^{\mu\nu} = \begin{bmatrix} E^2/c^2 & \frac{1}{c}(E_yB_z - E_zB_y) & \frac{1}{c}(E_zB_x - E_xB_z) & \frac{1}{c}(E_xB_y - E_yB_x) \\ \frac{1}{c}(E_yB_z - E_zB_y) & -\frac{1}{c^2}E_x^2 + B_z^2 + B_y^2 & -\frac{1}{c^2}E_xE_y - B_xB_y & -\frac{1}{c^2}E_xE_z - B_xB_z \\ \frac{1}{c}(E_zB_x - E_xB_z) & -\frac{1}{c^2}E_xE_y - B_xB_y & -\frac{1}{c^2}E_y^2 + B_x^2 + B_z^2 & -\frac{1}{c^2}E_yE_z - B_yB_z \\ \frac{1}{c}(E_xB_y - E_yB_x) & -\frac{1}{c^2}E_xE_z - B_xB_z & -\frac{1}{c^2}E_yE_z - B_yB_z & -\frac{1}{c^2}E_z^2 + B_x^2 + B_y^2 \end{bmatrix}$$

Figuur 1: two regions of the matrix $H^{\mu\nu}$

the green regions are given by the cross product $\frac{1}{c}(\vec{E} \times \vec{B})$, the red region is a 3x3 matrix given by the formula $-\frac{1}{c^2}E_iE_j + \delta_{ij}B^2 - B_iB_j$

Furthermore we have:

$$F_{\alpha\beta}F^{\alpha\beta} = 2(B^2 - \frac{1}{c^2}E^2) \tag{2}$$

and

$$\eta^{\mu\nu} = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3}$$

therefore, we can define a tensor $K^{\mu\nu} = \frac{1}{4}\eta^{\mu\nu}F_{\alpha\beta}F^{\alpha\beta}$ for which:

$$K^{\mu\nu} = \frac{1}{2}(B^2 - \frac{1}{c^2}E^2)\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4}$$

Now we can define

$$\mathbb{T}^{\mu\nu} = \frac{1}{\mu_0}(H^{\mu\nu} - K^{\mu\nu}) \tag{5}$$

1.1 electromagnetic energy density

In 1 we see that $H^{00} = E^2/c^2$ and in 4 that $K^{00} = -\frac{1}{2}(B^2 - \frac{1}{c^2}E^2)$, therefore we find that $\mathbb{T}^{00} = \frac{1}{2}(\varepsilon_0E^2 + \frac{1}{\mu_0}B^2)$, where we have used the fact that $\frac{1}{c^2} = \varepsilon_0\mu_0$

1.2 Poynting vector

$\mathbb{T}^{0i}$ is the ith component of a vector composed of the 1st, 2nd and 3rd elements of the 0th row of tensor $\mathbb{T}$. In 1, we see that this corresponds to the green region, thus the cross product $\frac{1}{c}(\vec{E} \times \vec{B})$. We know $S = \frac{1}{\mu_0}(\vec{E} \times \vec{B})$ from the handbook. We thus find indeed:

$$\mathbb{T}^{0i} = \frac{1}{c}(\vec{E} \times \vec{B})_i = \frac{1}{c\mu_0}\varepsilon_{ijk}E_jB_k = \frac{1}{c}S_i \tag{6}$$

1.3 Maxwell stress tensor

We know from 1 that:

$$H^{ij} = -\frac{1}{c^2}E_iE_j + \delta_{ij}B^2 - B_iB_j \tag{7}$$

Furthermore, we see from 4 that:

$$K^{ij} = \frac{1}{2}(B^2 - \frac{1}{c^2}E^2)\delta_{ij} \tag{8}$$

putting this together according to 5 and once again considering that $\frac{1}{c^2} = \varepsilon_0\mu_0$, we find indeed:

$$\mathbb{T}^{ij} = -\varepsilon_0(E_iE_j - \frac{1}{2}\delta_{ij}E^2) - \frac{1}{\mu_0}(B_iB_j - \frac{1}{2}\delta_{ij}B^2) = -T_{ij} \tag{9}$$

1.4 conservation of energy and momentum

First take the derivative of the 0th column:

$$\partial_\mu\mathbb{T}^{\mu 0} = \frac{1}{c}\frac{\partial}{\partial t}\mathbb{T}^{00} + \nabla\mathbb{T}^{i0} = 0 \tag{10}$$

since $\mathbb{T}^{i0} = \frac{1}{c}S_i$ and $\mathbb{T}^{00} = u$, we find:

$$-\nabla \cdot \vec{S} = \frac{\partial}{\partial t}u \tag{11}$$

which is the continuity equation of energy, according to p365 eq8.12 in the handbook

Now, take the partial derivatives for the 1st, 2nd and 3rd columns. Looking at 1 and 4 we can clearly see that we will have to combine the results of these equations. Moreover, the partial derivatives of the x^0 component, the temporal partial derivative, will combine into a component corresponding to $\frac{1}{c}(\vec{E} \times \vec{B})$, and the spacial partial derivatives will combine into a component corresponding to $-\varepsilon_0(E_iE_j - \frac{1}{2}\delta_{ij}E^2) - \frac{1}{\mu_0}(B_iB_j - \frac{1}{2}\delta_{ij}B^2)$ We thus see that:

$$\partial_\mu\mathbb{T}^{\mu i} = \frac{1}{c}\frac{\partial}{\partial t}(\frac{1}{c}(\vec{E} \times \vec{B})) + \nabla \cdot (-\varepsilon_0(E_iE_j - \frac{1}{2}\delta_{ij}E^2) - \frac{1}{\mu_0}(B_iB_j - \frac{1}{2}\delta_{ij}B^2)) = 0 \tag{12}$$

Of course, we know from 1.2 and 1.3 that we can simplify these into the Poyting vector and the Maxwell stress tensor, giving us the equation:

$$\begin{aligned}\nabla \cdot T_{ij} &= \frac{1}{c^2}\frac{\partial}{\partial t}\vec{S} \\ &= \varepsilon_0\mu_0\frac{\partial}{\partial t}\vec{S}\end{aligned}$$

according to p370 eq8.19 this means that **f**, the force per unit volume, is zero. If there is no force exerted on the system, the momentum will be conserved.

2 problem 2: Minkowski force

2.1 spatial component

We can write out:

$$q\eta_\nu F^{\mu\nu} = q\gamma\begin{pmatrix}\frac{1}{c}(u_xE_x + u_yE_y + u_zE_z) \\ E_x + B_zu_y - B_yu_z \\ E_y + B_xu_z - B_zu_x \\ E_z + B_yu_x - B_xu_y\end{pmatrix} \tag{13}$$

where we have to remember that $\eta_\nu = (-\gamma c, \gamma \mathbf{u})$, since we had to lower the index.
Furthermore, from p549 eq12.70 in the handbook we know:

$$\mathbf{K} = \gamma \mathbf{F} \tag{14}$$

thus we find indeed from the spatial part of 13 that:

$$\mathbf{K} = q\gamma(\vec{E} + \vec{u} \times \vec{B}) \tag{15}$$

and thus that:

$$\mathbf{F} = q(\vec{E} + \vec{u} \times \vec{B}) \tag{16}$$

2.2 zeroth component

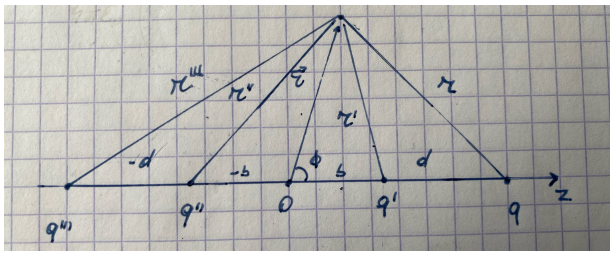
from 13 we can clearly deduce that

$$K^0 = \frac{q\gamma}{c}(\vec{u} \cdot \vec{E}) \tag{17}$$

3 problem 3: conductor

3.1 potential V

In the handbook chapter 3.2, we considered both the potential of an infinite grounded plane and of a conducting sphere. Combine both these examples into a new configuration. For the plate we want a symmetry around the z-plane, while for the sphere we want a charge $q' = -\frac{R}{d}q$ at a distance $b = \frac{R^2}{d}$. We thus decide to take a configuration with four charges. q at a distance d from the origin, q' at a distance b, $q'' = \frac{R}{d}q$ at a distance -b and $q''' = -q$ at a distance -d from the origin. This is shown in the figure:



Figuur 2: configuration

I will be using **r** for curly r, since i cant find the right latex package. We thus find for the potential V:

$$V = \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{\mathbf{r}} + \frac{q'}{\mathbf{r'}} + \frac{q''}{\mathbf{r''}} + \frac{q'''}{\mathbf{r'''}}\right) \tag{18}$$

Now we know for the configuration of the plate p124 eq3.9:

$$V_{plate}(x,y,z) = \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{\sqrt{x^2+y^2+(z-d)^2}} - \frac{q}{\sqrt{x^2+y^2+(z+d)^2}}\right) \tag{19}$$

in spherical coordinates, this becomes:

$$V_{plate}(r,\theta,\phi) = \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{\sqrt{r^2-2rdcos\phi+d^2}} - \frac{q}{\sqrt{r^2+2rdcos\phi+d^2}}\right) \tag{20}$$

furthermore, from the configuration of the sphere p127 eq3.19 we know that:

$$V_{sphere}(r,\theta,\phi) = \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{\sqrt{r^2-2rdcos\phi+d^2}} - \frac{q}{\sqrt{R^2-2rdcos\phi+(rd/R)^2}}\right) \tag{21}$$

Now we need to combine these in the right way. We have a superposition of the configuration of the plate and the sphere, but we also need to account for the charge q'' which is similar to the configuration of the sphere.
We can see that the point charge q contributes a term $\frac{q}{\sqrt{r^2-2rdcos\phi+d^2}}$, while q''' contributes a similar,

but slightly different term $-\frac{q}{\sqrt{r^2+2rdcos\phi+d^2}}$. q' contributes a term $-\frac{q}{\sqrt{R^2-2rdcos\phi+(rd/R)^2}}$, whilst once again similar but slightly different q'' contributes $\frac{q}{\sqrt{R^2+2rdcos\phi+(rd/R)^2}}$.

$$\begin{aligned} V(r,\theta,\phi) =& \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{\sqrt{r^2-2rdcos\phi+d^2}} - \frac{q}{\sqrt{r^2+2rdcos\phi+d^2}}\right. \\ & \left. - \frac{q}{\sqrt{R^2-2rdcos\phi+(rd/R)^2}} + \frac{q}{\sqrt{R^2+2rdcos\phi+(rd/R)^2}}\right) \end{aligned}$$

to check wether this is correct, we need to consider the boundary conditions. We need for r=R, that V=0:

$$\begin{aligned} V(R,\theta,\phi) &= \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{\sqrt{R^2-2Rdcos\phi+d^2}} - \frac{q}{\sqrt{R^2+2Rdcos\phi+d^2}} - \frac{q}{\sqrt{R^2-2Rdcos\phi+d^2}} + \frac{q}{\sqrt{R^2+2Rdcos\phi+R^2}}\right) \\ &= 0 \end{aligned}$$

since the first two and second two terms cancel out.
Furthermore, the potental should be zero at z=0, or thus at $cos\phi = 0$:

$$V(r,\theta,\pi/2) = \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{\sqrt{r^2+d^2}} - \frac{q}{\sqrt{r^2+d^2}} - \frac{q}{\sqrt{R^2+(rd/R)^2}} + \frac{q}{\sqrt{R^2+(rd/R)^2}}\right) = 0 \tag{22}$$

where once again the forst two and second two terms cancel out. We thus have found a potential that corresponds to the boundary conditions.

3.2 charge density

according to eq2.49:

$$\sigma = -\varepsilon_0 \frac{\partial V}{\partial n} \tag{23}$$

with $\frac{\partial V}{\partial n}$ the normal derivative of V at the surface. This means we need two derivatives, since we have two cases. First of all we have the normal on the sphere at r=R. Furthermore we have the normal on the surface of the plate. This is at $z = 0$ in carthesian coordinates:

$$\sigma = -\varepsilon_0\left(\frac{\partial V(x,y,z)}{\partial z}|_{z=0} + \frac{\partial V(r,\theta,\phi)}{\partial r}|_{r=R}\right) \tag{24}$$

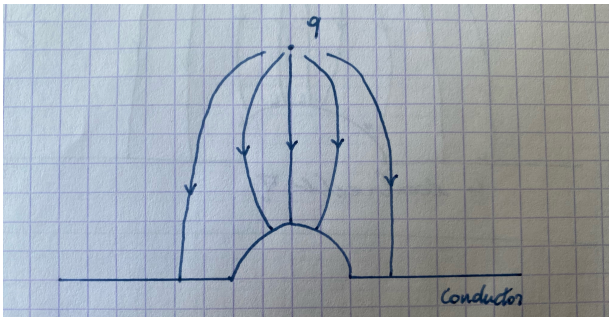
3.3 induced charge

we know from eq3.11 that the induced charge in the plate is -q. Analogously we can find for our setup that the induced charge is just the charge we considered for the 'trick' setup:

$$Q = q' + q'' + q''' = -\frac{R}{d}q + \frac{R}{d}q - q = -q \tag{25}$$

3.4 field lines

The electric field lines are always perpendicular to the surface of the conductor. It might not be so clear in the drawing, but the field lines are supposed to be perpendicular to the surface of the conductor. Furthermore, they go from a positive charge, in this case q, to a negative charge, in this case the conductor.



Figuur 3: field lines

4 problem 4: spherical shell

4.1 radial dependence of the potential

This problem can be tackled using separation of variables analogous to example 3.6 and 3.7 in the handbook. The most general solution is equation 3.65 on p141:

$$V(r,\theta) = \sum_{l=0}^{\infty}(A_l r^l + \frac{B_l}{r^{l+1}})P_l(cos\theta) \tag{26}$$

First consider the inside of the sphere:

4.1.1 inside

We don't want V to blow up at the origin, thus we need $B_l = 0$, thus:

$$V(r,\theta) = \sum_{l=0}^{\infty} A_l r^l P_l(cos\theta) \tag{27}$$

Now at r=R, thus should be equal to $V(\theta)$, thus:

$$V(R,\theta) = \sum_{l=0}^{\infty} A_l R^l P_l(cos\theta) = V_0 cos^2\theta \tag{28}$$

rewrite $V(\theta)$ as:

$$V(\theta) = \frac{2}{3}V_0(\frac{1}{2}(3cos^2\theta - 1)) + \frac{1}{3}V_0 \tag{29}$$

where we recognize the term in the parenthesis to be $P_2(cos\theta)$ and the second term can be written in terms of $P_0(cos\theta) = 1$

$$V(\theta) = \frac{2}{3}V_0 P_2(cos\theta) + \frac{1}{3}V_0 P_0(cos\theta) \tag{30}$$

therefore we find $A_0 = \frac{1}{3}V_0$ and $A_2 = \frac{2}{3}\frac{V_0}{R^2}$ whereas the other A_l are zero. Inside the sphere we thus have:

$$V(r,\theta) = \frac{1}{3}V_0(1 + \frac{r^2}{R^2}(3cos^2\theta - 1)) \tag{31}$$

4.1.2 outside

Since the potential needs to go to zero for $r \to \infty$, we have $A_l = 0$. We still have that:

$$V(\theta) = \frac{2}{3}V_0(\frac{1}{2}(3cos^2\theta - 1)) + \frac{1}{3}V_0 \tag{32}$$

thus we find $B_0 = \frac{1}{3}V_0 R$ and $B_2 = \frac{2}{3}V_0 R^3$, whereas the other B_l are zero. We thus have a potential outside the sphere:

$$V(r,\theta) = \frac{1}{3}V_0(\frac{R^3}{r^3}(3cos^2\theta - 1) + \frac{R}{r}) \tag{33}$$

4.2 dipole moment

We can calculate the induced surface charge density using formula 2.35 on p87 in the handbook:

$$\nabla V_{outside} - \nabla V_{inside} = -\frac{1}{\varepsilon_0}\hat{\mathbf{n}} \tag{34}$$

This means we only have to take the gradient of V_{inside} and $V_{outside}$ according to the direction normal to the sphere, which is the $\hat{r}$ direction. This means that we can rewrite the formula as:

$$\frac{\partial}{\partial r}V_{outside} - \frac{\partial}{\partial r}V_{inside} = -\frac{1}{\varepsilon_0} \tag{35}$$

where we can calculate:

$$\frac{\partial}{\partial r}V_{inside} = 2\frac{V_0}{R}cos^2\theta - \frac{2}{3}\frac{V_0}{R} \tag{36}$$

and:

$$\frac{\partial}{\partial r}V_{out} = \frac{2}{3}\frac{V_0}{R} - 3\frac{V_0}{R}cos^2\theta \tag{37}$$

therfore:

$$\begin{aligned} -\frac{1}{\varepsilon_0}\sigma &= \frac{4}{3}\frac{V_0}{R} - \frac{5V_0}{R}cos^2\theta \\ \Leftrightarrow \sigma &= \frac{V_0\varepsilon_0}{R}(5cos^2\theta - \frac{4}{3}) \end{aligned}$$

we can calculate the dipole moment via eq3.98 p151:

$$\mathbf{p} = \int \mathbf{r'}\rho(\mathbf{r'})d\tau' \tag{38}$$

where we have a charge density:

$$\rho(\mathbf{r'}) = \begin{cases} \sigma & r' = R \\ 0 & r' \neq R \end{cases} \tag{39}$$

then we have:

$$\begin{aligned} \mathbf{p} &= \int_0^\infty \int_0^{2\pi} \int_0^\pi r' \frac{V_0\varepsilon_0}{R}(5cos^2\theta - \frac{4}{3})r'^2 sin\theta dr' d\theta d\phi \\ &= R^3\frac{V_0\varepsilon_0}{R}\pi\left(5\int_0^{2\pi} cos^2\theta sin\theta d\theta - \frac{4}{3}\int_0^{2\pi} sin\theta d\theta\right) \\ &= V_0\varepsilon_0\pi R^2\left(\frac{5}{3}cos^3\theta + cos\theta\right)_0^{2\pi} \end{aligned}$$

where we find:

$$\left(\frac{5}{3}cos^3\theta + cos\theta\right)_0^{2\pi} = 0 \tag{40}$$

thus we have:

$$\mathbf{p} = 0 \tag{41}$$

which is pretty logical. We have a potential thats symmetric around the equator. Therefore this problem is similar to two charges an equal distance away from each other, which also produces a dipole moment of 0.

5 problem 5: multipoles

For this exercise we can use the results of problem 3.57 p163 in the handbook:

$$V_{mon} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r} \qquad V_{dip} = \frac{1}{4\pi\varepsilon_0}\frac{\sum \hat{r}_i p_i}{r^2} \qquad V_{quad} = \frac{1}{4\pi\varepsilon_0}\frac{\sum \hat{r}_i\hat{r}_j Q_{ij}}{r^3} \tag{42}$$

Now, the charges all lie on x=y=0 with differing heights z. We thus have a charge density ρ:

$$\rho(\mathbf{r'}) = \begin{cases} -q & z = \pm a, x = y = 0 \\ 2q & z = 0, x = y = 0 \\ 0 & everywhere \quad else \end{cases} \tag{43}$$

5.1 monopole

we have:

$$p_0 = -q + 2q - q = 0 \tag{44}$$

we thus find a monopole moment $p_0 = 0$

5.2 dipole

we now have according to eq3.100:

$$\sum \mathbf{r'}_i q_i = -qa + 2q \cdot 0 + qa = 0 \tag{45}$$

thus we find once again:

$$p_1 = 0 \tag{46}$$

5.3 quadrupole

we have:

$$Q_{ij} = \frac{1}{2} \int [3r'_i r'_j - (r')^2 \delta_{ij}] \rho(\mathbf{r'}) d\tau \tag{47}$$

since we have a charge density:

$$\rho(\mathbf{r'}) = \begin{cases} -q & z = \pm a, x = y = 0 \\ 2q & z = 0, x = y = 0 \\ 0 & everywhere \quad else \end{cases} \tag{48}$$

we can calculate this as:

$$Q_{ij} = \begin{bmatrix} qa^2 & 0 & 0 \\ 0 & qa^2 & 0 \\ 0 & 0 & -2qa^2 \end{bmatrix} \tag{49}$$